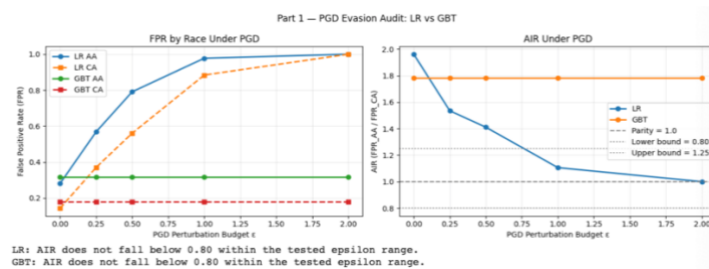


## Individual Homework 05 - Natchapa Aunkay (G27589366)

### Part 1: PGD Evasion Audit

Under PGD evasion, the Logistic Regression (LR) model is highly sensitive to perturbations. As  $\epsilon$  increases, the false positive rate (FPR) for both African-American and Caucasian groups rises sharply, approaching 1.0. This indicates that the model increasingly classifies nearly all individuals as high risk. Although the Adverse Impact Ratio (AIR) moves toward 1.0 (parity), this does not reflect improved fairness but rather uniformly poor predictions across groups.

In contrast, the Gradient Boosted Tree (GBT) model remains largely unchanged across all  $\epsilon$  values. Both FPR and AIR stay stable, suggesting that the attack has little effect on its predictions. This apparent robustness is likely due to the structure of tree-based models, which are less responsive to small gradient-based perturbations.



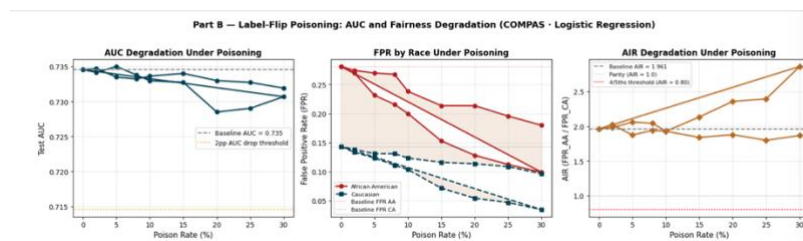
Overall, the results show that LR is significantly more vulnerable to PGD attacks than GBT. This implies that model selection in high-stakes applications should consider not only predictive performance but also robustness to adversarial manipulation.

### Part 2: Poisoning Loop with Fairness Monitoring

Across all poison rates, AUC remains stable (within ~2 percentage points), indicating that standard performance monitoring would not detect the attack. However, fairness metrics change substantially. When targeting African-American defendants, AIR shifts significantly, indicating that fairness can be manipulated depending on the targeted group. When targeting Caucasian defendants, AIR decreases slightly.

This demonstrates a stealth attack: model accuracy appears unchanged while fairness degrades. The effect is visible even at low poison rates, where AIR moves outside the acceptable range [0.80, 1.25]. Traditional PSI-based drift detection would likely fail, since the attack modifies labels rather than feature distributions.

- **Figure 2:** Poisoning — AUC, FPR, and AIR degradation curves



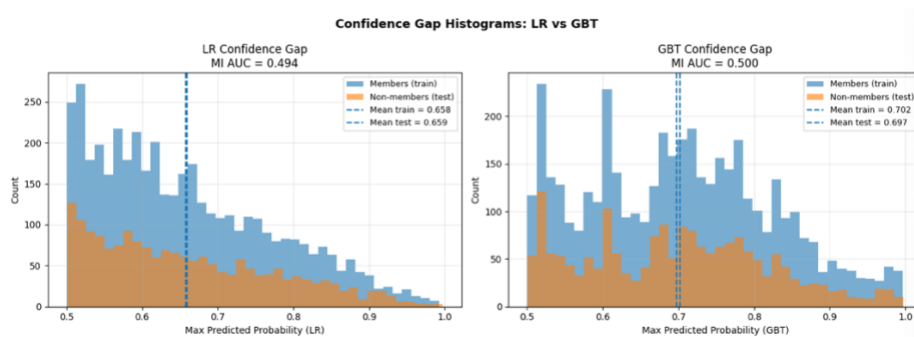
A PSI-based drift monitor would likely fail to detect this attack because poisoning modifies training labels rather than feature distributions. This highlights a critical limitation of traditional monitoring systems and underscores the importance of fairness-aware evaluation.

### **Part 3: Membership Inference**

Both LR and GBT achieve MI AUC values near 0.50, indicating the attack performs no better than random guessing. The confidence-gap histograms show strong overlap between training and test samples, with nearly identical mean confidence.

Although GBT has a slightly larger generalization gap than LR, this does not translate into higher MI risk. Overall, privacy leakage under this attack appears low, though stronger attacks may yield different results.

- **Figure 3: Membership Inference — Confidence gap histograms (LR vs GBT)**



Overall, privacy risk under this attack appears low, though stronger attacks may yield different results.

### **Part 4: Reflection**

The highest-risk finding across all experiments is the presence of a stealth poisoning vulnerability, where fairness can be significantly degraded without noticeable changes in performance. This represents a critical risk in high-stakes systems, as decision-makers may trust a model that appears accurate while it systematically produces biased outcomes.

A proactive mitigation strategy is to implement fairness-aware monitoring by tracking metrics such as AIR alongside AUC during model evaluation. In this study, AIR changed significantly while AUC remained stable, demonstrating that performance metrics alone are insufficient. A reactive mitigation strategy is to introduce data integrity controls, such as auditing training labels, maintaining secure data pipelines, and applying anomaly detection to identify suspicious samples.

These mitigations introduce important tradeoffs. Enforcing fairness constraints may reduce predictive performance, while increased monitoring adds operational complexity and cost. Additionally, improving fairness for one group may unintentionally increase errors for another group, highlighting the inherent tension between accuracy, fairness, and robustness.

These findings highlight that robustness, fairness, and privacy must be evaluated jointly, as optimizing one dimension alone can mask risks in others.